Impulse changes momentum

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect force over time to momentum change.

SUCCESS CRITERIA - I can...

1. Calculate impulse from force and duration.
2. Match impulse direction to momentum change.
3. Use impulse to find a final velocity.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “impulse” and “average” to explain your model or result. Sentence starters are optional.

impulse: net force acting over time; change in momentum

average: one value representing an interval or a set of values

momentum: mass times velocity, including direction

01 Warm-up: bring back an earlier idea

An ideal circular orbit has GM=225 m^3/s^2 and r=9 m.
Find speed, inward acceleration and period; explain how velocity changes.

Word help: ideal = using stated simplifying conditions

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

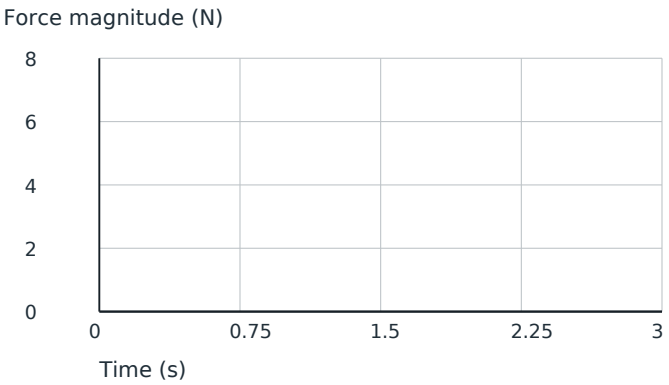
02 Worked example: explain the model

Reference: $J = F_{avg}\,\Delta t = \Delta p = p_f - p_i$.
J is impulse; F_{avg} is average net force. Signed force-time area gives impulse; i = initial, f = final.

Worked reasoning: A constant 6 N force acts right for 2 s. Impulse is 12 N·s right, so momentum increases by 12 kg·m/s.

Your turn: Draw the force-time rectangle and shade its area.
Explain why the two impulse units are equivalent.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



Marzano check - circle: ① ② ③

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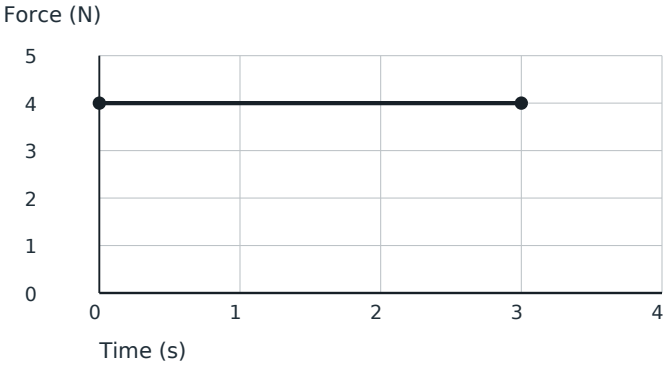
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Read the rectangular force-time graph.
Find the area and state what it means.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3

04 Apply the physics

A 0.50 kg ball hits a wall while moving at +8 m/s and rebounds at -6 m/s. Contact lasts 0.020 s.
Label the before-and-after model.
Calculate impulse and average net force on the ball.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Before	**After**

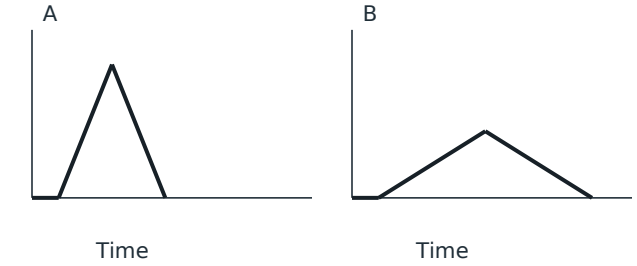
Label mass and velocity.

Marzano check - circle: 1 2 3

05 Explain your reasoning

Two force pulses have equal signed area but different peak heights. What must be equal? What need not be equal?

Sentence starter: Both ... . They differ in ... because ... .



Same scales: compare impulse and peak force.

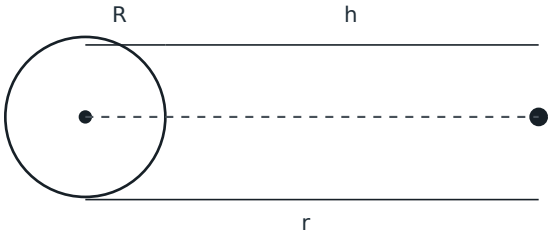
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

At what altitude h/R is a spherical planet's field 1/16 of its surface value? Repeat for 1/36.
Label the masses, field or force relevant to the question.

Word help: altitude = height above a reference surface

Sentence starter: My design shows ... . It meets the requirement by ... .



Marzano check - circle: 1 2 3

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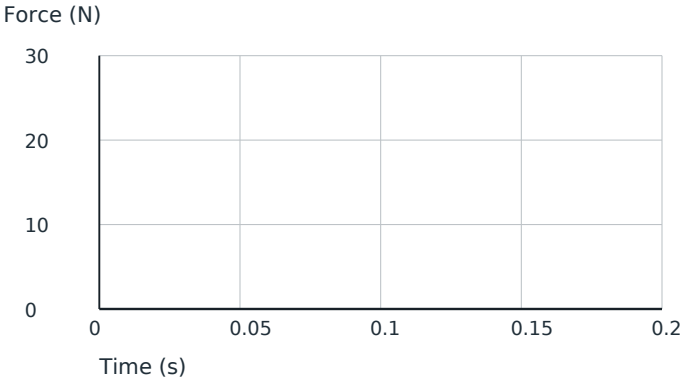
Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A triangular force pulse has base 0.20 s and peak 30 N.
Find impulse and velocity change for a 0.60 kg object.

Sentence starter: I found ... [unit]. This value means ... in this situation.

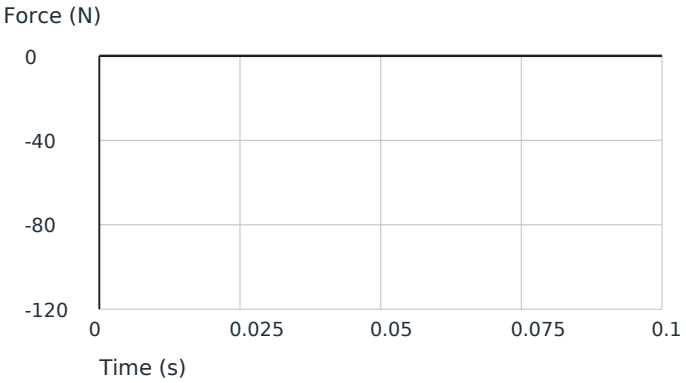


Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 2 kg cart changes velocity from +3 to -2 m/s in 0.1 s.
Find impulse and average force.
Sketch a force-time rectangle with the correct signed area.

Sentence starter: I found … [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3